H48W35 - Software Development: Graded Unit 2

Speedo Models Web System Planning Report

Stage 1.1.1

Initial Planning-

An analysis of the project assignment brief.

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Introduction to Project

Aims of the Project

1.1.1.1 Aims of the Project assignment

The goal of the project is to build a computerised system for “Speedo Models” a family run business. This website system aims to tackle the inefficiencies and problems that comes with running a paper-based process of ordering and purchasing.
The building of this system will be broken down into three stages:

Stage one is the planning stage. For this stage I will create a report – I will analyse the project brief, ask the client questions, write a requirement specification sheet, build initial UML diagrams – class and use case diagrams, identifying and describing the resources and materials that will be used and identify the information source. I will also create a formal project plan to keep track of the progress being made, including main milestones, a breakdown of tasks per stage, realistic timescales, scheduling of tasks, deliverable and resources used and following a project deadline.
The purpose of all of this is for me to have a clear vision of what I am building. Then I will move on to the design stage where I will be building more detailed UML diagrams (class, use case, activity, sequence diagrams), I will normalize the data provided and then build entity relationship diagrams with the normalized data. I will create Event Handling forms, then a module descriptor followed by a complete screen layout charts document before moving onto the implementation stage.

Stage two will include the development of the application, the creation of testing documents to ensure everything is working, building external documentation for the user to follow, and updating the project plan as I work on this stage. The purpose of this stage is to build a functional application, that will suit the customer's needs and to keep track of the progress and any bugs in the application.

Stage three is to write an evaluation report that covers the outline of the assignment, the strengths and weaknesses of the project, any modifications that took place, knowledge and skills gained, and any recommendations I would make for any future development

projects. The purpose of this stage is to arrive at a conclusion for the project and through reflection of the process, highlight how I've grown as a developer.

Review the Project plan to see breakdown of each task per stage.

Overview of Project Brief

1.1.1.2 Interpretation of the project assignment brief

The objective of the project is to transform the Speedo models paper-based order and purchasing system to a streamlined, computerized e-commerce website. This website will have efficient online ordering for customers and a back-end management function for employees. This update will address inefficiencies in the current system and reduce human error.

Initial High-Level Requirements:

Interface – Website System

Functional Requirements-

Users and how they interact with Products/Orders -

The system will include the following users who will be able to interact with the system: Unregistered users who can view/browse products, filter them by category, select an item to purchase, login or register and interact with all the other elements of website: the contact page and about us page. Registered users can do everything the unregistered user can do and have a specific role with key functionalities these include:

- Employees – Who have access to the backend of the system and can add/amend products (product management), generate reports, view, amend and confirm customer orders (order management). Specialized roles include:
    - Admin
    - Sales assistant
    - Invoice Clerk
    - Sales Manager
    - Assistant Manager
    - Stock Control Manager
    - Warehouse Assistant
- Customers – Who can place orders, make payments, amend/cancel their own orders, fill out shipping details, track an order and print a return slip.

The system will also hold all registered user information such as name, address, etc (for more information please refer to the *requirements specifications* section of the report).

Other System Requirements -

The system will send invoices to customers after they make a purchase.
The system will have a loyalty program which will offer discounts for premium customers.
The system will have automatic alerts for customers and staff when the stock of an item is low and a section on the contact us page for querying stock availability.
Integration of payment gateways will include credit card and pay pal.
Product, customer, and staff creation all fall to the staff members (only the admin can create other staff).
Customers can terminate a website visit, without placing an order – which removes all details of their order from the system and the stock levels are left unchanged.
Specific staff can increase stock levels and change product price (view functional requirements specification document section of this report).
Customers can get a refund on their order, if they are unsatisfied.

System Interface-

There will be a defined error message for various actions such as login failures, registration issues and product/creation/modification errors.
In terms of authentication, there will be usernames and passwords for each registered user, character and minimum length restrictions will be put in place.

Non-Functional Requirements-

The system must be secured, user-friendly and compatible across various devices and browsers. The design should incorporate Speedo Models branding and be visually appealing.

The system will have to comply with the GDPR. The system also must be compatible with widespread legacy and contemporary desktop browsers.
In-terms of security there will be measures such as data encryption, secure payment gateways (mentioned earlier) and user authentication & authorization protocols that will be implemented.

Good thorough internal documentation of the code (detailed commenting) – for improved code readability and in support of best programming practices.

Initial Investigation

Fact Finding Techniques

1.1.1.3 Information gathered to clarify brief.

Client Interview Questions:

The way I gathered information, was through utilizing the following techniques: I conducted *client interview questions* by submitting a form with questions, which the client answered in their own time. With this technique I was able to clarify the given project brief and asked questions which filled in the information gaps.

The interview questions helped clarify the client's needs and expectations; as well as reducing human error and enhance operation efficiency, the client also wants to expand their market reach, they expressed a desire to maintain their current product range, whilst increasing their reach through online sales. They highlighted the importance of a user-friendly e-commerce platform that supports all the requirements specifications (mentioned in more detail later).

I sent the interview questions to the client and waited for their response before moving forward. Another approach is to interview in person, although there are drawbacks to this, a few include: it being more time-consuming and expensive (due to traveling to the interview location), you might get resistance due to the client not understanding the system, fear of not being able to cope with the change, or the user underestimating the cost of specific functionalities of the system and project scope creep, *this information was **cited** from a LinkedIn Article by Charlie Marshall (2016).*

Observation:

Another fact-finding technique that wasn't used, but I would consider the next time I take on a similar project would be *observation,* which involves an analyst directly

watching the end-users or system in action, with the end user being aware of this observation.
This technique is non-interactive: the analyst watches the process without interrupting the workflow, which allows the analyst to act as a real-world user and look at the system in depth, enabling the analyst to uncover issues not reported by the user, which they can then offer insights into potential improvements and solutions for the application.

A limit to this technique is the effectiveness of the observation, as it relies on the analyst's ability to interpret what they see correctly.
Another disadvantage is the Hawthorne Effect which states people may alter their behaviour because they know they are being observed. This will affect results.

Questionnaires:
In addition to the techniques discussed above another well-known one is questionnaires, which are utilized to gather data from a large group of people, and we can compare the collected data to a competitor's website and adjust appropriately.
In this use case, we could have used it to compare Speedo models to its main competitors, focusing on website functionality, product range and customer experience. By selecting a group of potential users to undertake a questionnaire, we would have been able to collect valuable feedback on what customers value in a car model collection retail website, and this would help us understand the functionality required to build a user-friendly e-commerce platform that also supports all the requirements specifications, ensuring that the website meets both the expectations and preferences of its target audience.

This way of gathering data allows us to identify areas where Speedo Models can differentiate itself from competitors such as RacingModels.com (the main competitor). Following a structured format, questionnaires can pinpoint specific features and services (that can potentially be added to the requirement specification document) that could enhance the business's competitive edge. This technique is similar to client-interview questions, but instead of asking the business, we are asking the end-user and potential customers directly, and thus is a powerful tool to use in conjunction with the 'Interview Question' technique.

The downside to this technique is that the process itself is time-consuming and expensive. Certain questions may be relevant to some users and irrelevant to others or some users may underestimate the cost of a specific functionality that they would like to see as part of the system.

Other Fact-Finding Techniques that can be used:

- *Market research* – To identify current trends in e-commerce platforms for competitor businesses.
- *Legal Requirements* – reviewing relevant legislation, including GDPR for data protection and e-commerce regulations (link in appendices).
- *User Experience Best Practices* – this ensures that the website offers an intuitive and engaging shopping experience.

Client Interview Questions vs Observations:

Depending on the information you want to gather and the way you go about doing so, both techniques discussed offer different ways to achieve this and therefore have contrasting strengths and weaknesses as a result.

For example, gathering information via interviews means you are taking a direct approach by asking questions and gathering answers, while observation indirectly gathers information by watching how tasks are performed.

In interviews you interact with the client by asking targeted questions to gather specific information, wants, and needs, whereas with observation, you are not interacting with anyone, instead, the analyst acts as a passive viewer, in an attempt to avoid influencing user behaviour and thus offers a broader view of operational practices carried out and inefficiencies which affects user interaction, which might not be possible to uncover during an interview.

As observed, both techniques are vastly different – an interview can uncover the "why" behind user actions and decisions, whereas observation can reveal the "how" of daily operations and user interaction – using both techniques in conjunction with one another, generates a very powerful tool to gather data, which can be used to meet the clients goals when it comes to providing an efficient and effective website application that is user-friendly, and gives the client a competitive edge in the market.

Competitor Analysis:

As identified by the client, RacingModels.com serves as a benchmark for Speedo Models. The competitor has established a strong online presence with a focus on a good range of racing car models and accessories. Their website features include product categorization, detailed product information (technical specifications of products), and secure checkout processes, all of which contribute to a seamless customer experience. Speedo model aims to have more accessibility and a wider product range than their competitor to enhance brand visibility and customer engagement. With an integrated loyalty points system, the brand aims to reward its customers for choosing to purchase from Speedo Models.

Interview Question and Responses

Background Questions

1. What other products does your company sell in-store?

A. Apart from model Scalextric racing cars, Speedo Models may also sell accessories like tracks, controllers, and other related items.

2. What country are your based in?

A. Speedo Models was opened in 2002 in a small unit in Milton Keynes. We sell model Scalextrix racing cars to our customers via a catalogue. Most of our customers order by phone, however some do order from our premises. With over 20,000 models in stock for immediate dispatch, making Speedo Models one of the top retailers of scale models. We have the resources and facilities to offer an outstanding service. This is down to the experience of our staff and ability to source hard to obtain items. We ship products 5 days a week and very often same day depending on order times. We have built an enviable reputation based on being able to deliver immediately from our own stock. We are well established and have served over 5.000 collectors.
CONTACT INFORMATION: ADDRESS: Unit 3B Pineham Farm, Haversham, MK19 7DP England UK Tel no: 01908 225650 EMAIL info@speedomodels.com WORKING DAYS/HOURS: Mon - Fri / 8:00 AM - 4:00 PM

3. Do you want to be or are you already international?

A. Currently, Speedo Models operates locally, but there might be plans for international expansion in the future.

4. Would you like to sell the same type of products online or are there limitations?

A. The intention is to sell similar products online to cater to a wider audience, with the possibility of adding other products in the future.

5. Who is your target audience/ main user base?

A. Speedo Models targets hobbyists, collectors, and enthusiasts interested in model Scalextric racing cars.

6. Who are the main competitors and what do you like and/or dislike about their website?

A. We strive to become UK's largest racing cars models provider. Therefore, we are looking not only to have an online presence for selling different brands of car models, but also to expand the business and provide our customers with accessories and spare parts and a car models repair service.
Currently our main competitor is https://www.racingmodels.com/index.php?route=common/home

7. Are there any existing systems/web applications that would need integrating with?

A. The current system is outdated and inefficient.

8. Do you have any preferred technologies/frameworks?

A. We await your advice as an IT specialist.

Timeline and Budget Questions

9. What is your budget range for the project?
10. What is the specific timeline and the deadline for the project?

Please find below an extract of our contract for your consideration.

The Developer shall complete the development of the Software according to the milestones described on the form below:

Project Start Date		01/02/2024	Project End Date
Stages		Deliverables	Deadlines
1	Planning	Interview Questions	11/02/2024
		Planning Report	10/03/2024
		Initial Project Plan	10/03/2024
		Analysis and Design Report	24/03/2024
2	Development	Final Application	02/06/2024
		Testing Documentation	02/06/2024
		External Documentation	04/06/2024
		Updated Gantt Chart/Project Plan	04/06/2024
3	Evaluation	Evaluation Report	10/06/2024

In accordance with such milestones, the final product shall be delivered to the Client by 10/06/2024 (the "**Delivery Date**").

For a period of 6 months after delivery of the final product, the Developer shall provide the Client attention to answer any questions or assist solving any problems regarding the operation of the Software up to 40 hours free of charge and billed to the Client at a rate of £100 per hour for any assistance thereafter. The Developer agrees to respond to any reasonable request for assistance made by the Client regarding the Software within 10 days of the request.

The Developer shall provide to the Client after the Delivery Date, a cumulative 8 hours of training with respect to the operation of the Software if requested by the Client.

If the Software as delivered does not conform to the Specifications, the Client shall within 6 months of the Delivery Date notify the Developer in writing of the ways in which it does not conform with the Specifications. The Developer agrees that upon receiving such notice, it shall make reasonable efforts to correct any non-conformity.

In consideration for the Service, the Client shall pay the Company at the rate of £100 per hour (the "Hourly Rate"), with a maximum total fee for all work under this Software Development Agreement of £15,000. Fees billed under the Hourly Rate shall be due and payable upon the Developer providing the Client with an invoice. Invoices will be provided for work completed by the developer once every 4 weeks.

The Parties acknowledge and agree that the Client will hold all intellectual property rights in the Software including, but not limited to, copyright and trademark rights. The Developer agrees not to claim any such ownership in the Software's intellectual property at any time prior to or after the completion and delivery of the Software to the Client. The Software will not violate the intellectual property rights of any other party.

Justification Questions

11. What are your expectations you have for the web application and how will these expectations contribute to the success of your business?

A. The expectations include streamlining ordering processes, reducing errors, improving customer experience, and expanding market reach, ultimately contributing to increased sales and customer satisfaction.
What is required is to produce an IT solution that accomplishes the requirements below: • Register a new customer. • Enter a new product into the system. • Increase the stock of an existing item. • Change the price of a product. • View order details. • An alert system to identify low stock items. • Amending/cancelling customer orders • Customers querying about stock availability We also require a website that will allow customers to: • Track an order. • Place an order. • Make a Payment • Cancel an Order • Generate Returns Slip

12. What competitive advantage will having a web application help you achieve?

A. Having a web application can provide a competitive advantage by offering convenience, accessibility, and a wider product range compared to competitors. It also enhances brand visibility and customer engagement. The new system should have:

13. Will there be any coupons/discounts you offer loyal customers?

- If so are there loyalty points the system will have to account for?

> A. Yes, there might be loyalty programs, coupons, or discounts for loyal customers. The system needs to account for loyalty points and bonuses based on purchase history.

Business Rules Questions

14. When do you want the first iteration of the web application to be up a running?

A. 25th April 2024.

15. What are the primary business goals you aim to achieve with this web application?
    - Can you prioritize these objectives in terms of importance?

> 1. A database system that incorporates:
    - Customer details
    - Customer order history
    - Stock held
    - Product details
2. A website that allows users to:
    - Register a new account
    - Place orders
    - View existing orders
    - View products
    - Select items to purchase
    - Login
3. Functionality for staff to:
    - Register a new customer
    - Record a new product
    - Record an increase in the stock level of an existing product
    - Change the price of a product
    - View details of orders placed by a customer
    - Generating daily, weekly, monthly and yearly reports

Maintenance and Support Questions

16. What are your expectations when it comes to ongoing maintenance and support for the application?

> A. The expectation is to have ongoing maintenance and support for the application to ensure its smooth operation, address any issues, and implement updates or improvements as needed.

17. Is there an internal team that will be involved in the project?

A. There may be an internal team involved in the project, including stakeholders, managers, and IT personnel, depending on the scale and complexity of the project.

Success Metrics

18. How will you measure the success of the web application?

A. Success can be measured through various metrics such as increased sales, customer satisfaction ratings, website traffic, conversion rates, and repeat purchases.

19. Are there any specific key performance indicators you want to focus on?

A. Key performance indicators may include order volume, average order value, customer retention rate, website bounce rate, and conversion rate.

20. Will we get feedback data from customers to measure customer satisfaction, i.e. conducting an online survey?

A. Yes, collecting feedback data from customers, such as conducting online surveys, can help measure customer satisfaction and identify areas for improvement.

Project Scope Questions

21. How is the company currently fulfilling orders now?

A. Currently, orders are fulfilled through catalogue-based ordering processes.

22. How is this expected to be implemented into the new system?
    - Are there different constraints to consider on the software system that you didn't have to consider when selling via catalogue?

A. The new system will streamline and automate order processing, offering additional features like online ordering, customer registration, and order tracking. Constraints unique to software systems, such as data security and scalability, need to be considered.

System Questions

23. Is there a specific system you wish to be built (a stand-alone or web system)?

A. The preferred system could be a web-based application integrated with an e-commerce platform for online sales.

24. Does a system already exist?

A. Currently, there might not be an existing system, or if there is, it may be a manual paper-based system.

Non-Functional:

25. What are the legislations (if any) you would want users to comply with before registering?

A. The system needs to comply with relevant legislations such as data protection laws (e.g., GDPR) and online trading regulations.

26. If it is a website system, do you want the site to maintain compatibility with widespread legacy and contemporary desktop browsers?

A. Yes, the website should maintain compatibility with a range of desktop browsers to ensure accessibility for users.

27. Is there a specific design, in terms of colours, theme's you would like to follow?

A. Design preferences might include a user-friendly interface, responsive layout, and visually appealing graphics consistent with the company's branding.

28. Are there specific company logo/pictures you would like to be included in the design?

A. Company logos, images, and branding elements should be incorporated into the design for brand recognition. We would like to re-brand ourselves.

29. Do you want the system to be compatible with any specific devices?

A. The system should be compatible with various devices, including desktops, laptops, tablets, and smartphones, to cater to diverse user preferences.

30. Are all sales recorded by the company?

A. Yes, all sales should be recorded for inventory management, financial reporting, and customer service purposes.

31. Are there specific security measures that need to be put in place?

A. Security measures such as data encryption, secure payment gateways, and user authentication protocols need to be implemented to protect sensitive information.

Functional:

32. To clarify will the system require a search functionality, whether that be a managerial role searching for a specific user or any user searching for a product?

A. Yes, the system should include search functionality for users to find specific products or for staff to search for customer/order details.

33. If more than one role exists, for example, can a manager/admin promote a staff?

A. Yes, different roles such as managers and administrators may have privileges to perform specific actions like promoting staff or managing inventory.
We require you as the IT expert to develop an IT system that is secure. • Admin will have the responsibility to register new members of staff and provide them with a username (consists of first name, second name and company name for the domain name) and password • Sales Assistant – responsible for processing customer orders, entering of orders, taking payment, customer registration. • Invoices Clerk – responsible for sending out invoices to corporate customers and for the processing of payments linked to these orders. • Sale Manager – access to all data • Assistant Manager – same as Sales Assistant plus price amendments and stock control. • Stock Control Manager- responsible for stock control • Warehouse Assistants – responsible for the dispatch of goods to customers.

All users require a User Id and Password.

34. Are there specific payment gateways, that can or cannot be implemented?

A. The system should support various payment gateways such as credit/debit cards, PayPal, and other secure online payment methods.

35. Should different delivery options be implemented?

A. Yes, different delivery options such as standard shipping, express delivery, or click and collect should be implemented to cater to customer preferences.
• First Class • Second Class • Courier for Next Day delivery, we have several firms we use. The checkout facility automatically calculates shipping costs depending on customer location and includes them in the total price. All goods are dispatched from our warehouse.

36. Does the company have a social presence and would you like that to be implemented into the system?

A. Yes, social media integration can be implemented to enhance brand visibility, customer engagement, and marketing efforts.

37. Can users register/login using google/facebook or any other types of integrations?

A. Yes, users may be able to register/login using their existing Google or Facebook accounts for convenience.

Product Functionality Questions

38. Regarding the Scalextric racing cars you intend on selling, what are the item specifications for each product?

A. Detailed specifications for each product, including dimensions, materials, and compatibility, need to be provided.

The following details are held about products:

- Product Id,
- Product Name
- Product Description,
- Product Dimensions,
- Product Materials,
- Product Compatibility,
- Product Colour,
- Product Price,
- Quantity on hand
- Re-order quantity
- Category Id
- Supplier Id

39. If more products are being sold, will there be different categories for products, and if so, what will these categories be?

A. Yes, products should be organized into different categories (e.g., cars, tracks, accessories) for easier navigation and browsing.

40. How should products be displayed/categorised?

A. Products should be displayed in a visually appealing manner with clear categories, descriptions, and images to aid decision-making.

41. Which user has the right to create/add new products to the system?

A. Staff members with appropriate permissions should be able to add, edit, or remove products from the system (see roles below).

42. What is the price range for each product (if there is more than one)

A. Price ranges for each product category should be defined to help customers filter and search for products within their budget.

Staff Functionalities Questions

43. Will different staff members have different roles such as managers and non-managers?

A. Yes, different staff members may have different roles such as managers, sales assistants, and warehouse staff, each with specific responsibilities.

We require you as the IT expert to develop an IT system that is secure.

- Admin will have the responsibility to register new members of staff and provide them with a username (consists of first name, second name and company name for the domain name) and password.
- Sales Assistant – responsible for processing customer orders, entering of orders, taking payment, customer registration.
- Invoices Clerk – responsible for sending out invoices to corporate customers and for the processing of payments linked to these orders.
- Sale Manager – access to all data
- Assistant Manager – same as Sales Assistant plus price amendments and stock control.
- Stock Control Manager- responsible for stock control
- Warehouse Assistants – responsible for the dispatch of goods to customers.

Staff should be able to generate daily, weekly, monthly and yearly reports based on the needs of the business and their role permissions. The following reports should be available:

Stock Item/Stock Levels Report:

- Product Id,
- Product Name
- Product Category
- Product Colour (if appropriate/required),
- Product Price,
- Quantity on hand
- Re-order quantity
- Delivery lead times
- Supplier Id

Stock by Supplier Report:

- Supplier Id, then grouped by Product Id showing:
- Product Description,
- Product Colour (if appropriate/required),
- Product Price,
- Quantity on hand

- Re-order quantity
- Delivery lead times

Invoices Report:

- Invoice No
- Date of Invoice
- Invoice Amount
- Customer Id
- Overall totals by customer/invoice

Customer & Orders:

- Order No
- Date of Invoice
- Order Amount
- Customer Id
- Overall totals by customer/order

All reports should be able to be viewed on screen, with the option to print and also be able to be exported into Ms Excel or PDF.

Staff Information Questions

44. Will the system hold the following information about the staff:
    - Staff id/number?
    - Address?
    - Telephone number?
    - Salay information?
    - Job types?

A. The system may store staff information such as staff ID/number, contact details, job roles, and salary information for payroll and management purposes.

45. Staff can create new customers, but who will create new staff accounts in the system?

A. Staff accounts may be created by administrators or managers with appropriate permissions.

Order Functionality Questions

46. Before placing the order:
    - Can users remove products from their basket?
    - Can users update their order, either by adding a different product, or changing the quantity of a product from their basket?

A. Yes, users should be able to modify their orders before finalizing them, including adding or removing products and updating quantities.

47. Can only specific users place orders?

A. Registered users should be able to place orders, with specific criteria or restrictions if necessary. Also, staff can place orders on customers behalf when in store or over the phone.

48. Can only staff view all the order details of each customer?

A. No, customers should be able to view their orders, track orders. Also, staff members with appropriate permissions should be able to view all order details, including order history and individual order items.

Order Information Questions

49. Who can place and cancel orders?

A. For ordering online customers should be able to browse, view our products and check stock availability without the need to login in. We would like the following sequence of events when placing an order online:
• The customer will select their products and enter quantities required. • If the product is in stock, they will then submit their order and at this point will be required to submit their email address and password as part of the customer verification process. If a customer has not registered previously then they will have to prior to placing their order. • When the customer submits their order via the website, the stock figure will be adjusted automatically.

- An email and SMS message will be send to the customer to confirm their order.

If a customer has not registered previously then they will have to prior to placing their order:

- enter their details: first name, last name, full address including their post code, telephone number and submit these details along with their email address and their chosen password.
- they will then receive a confirmation email asking them to click on a link to activate their account, once this has been done, they can then submit their order.

Once the order is placed, our warehouse assistants will process the order and get it ready for delivery. It is the warehouse assistant's responsibility to update the order status. Once the order has been delivered the status should be updated.

Also, customers will log in and be able to see all their orders (past & current) or they can use the tracking number that will be provided in the email confirmation of their order and check the order status.

A SMS message should be sent to the customer to inform about the delivery of their order.

50. Are there restrictions on cancelling orders?

A. Orders can be cancelled from 48 hours after order placement.
When a customer cancels an order we check the delivery date, if it is less than 48 hours before the delivery date there is a 25% cancellation fee. Orders can be cancelled up to 48 hours after order placement. All details regarding a cancelled order require to be deleted from the system.

Orders can be amended online up to 48 hours after order placement. If it is less than 48 hours prior to the delivery date, there will be a 10% charge. Order details including delivery can be amended online up to 48 hours after order placement. If it is less than 48 hours prior to the delivery date the change to delivery method may not be able to be accommodated.

51. Are there any product return policies the company must abide by?

A. Customers can return items within 28 days of purchase.
This can be done online, by accessing their order and printing out a returns label and mailing the item to us or alternatively by bringing the item(s) into the store. • We require that the following details are recorded for each item returned: • Order Id • Item Code • Date of Return • Reason for return • Condition of item • Customer No • Customer contact details When an order is returned, then the stock manager will re-stock the item if the product is in good working order or return it to the supplier if it's faulty.

Functional and Non-Functional Requirements

1.1.1.4 Initial functional and non-functional requirements and constraints

SOFTWARE ENGINEERING DOCUMENTATION REQUIREMENTS SPECIFICATION	LEVEL OF PRIORITY FROM HIGHEST TO LOWEST
[redacted]	

GENERAL REQUIREMENTS OF THE SYSTEM	IMPLEMENTATION PRIORTY

Functional Requirements:
• The system must allow all company staff to:
  ○ Record a new product.
  ○ Generate daily, weekly, monthly, and yearly reports.
  ○ View details of orders placed by a customer.
  ○ Amend/Cancel & refund a customer order.
• The system must allow customers to:
  ○ Place orders.
  ○ Make a payment.
  ○ View existing orders.
  ○ Amend/cancel orders.
  ○ Track an order.
  ○ Print returns slip.
• The system will allow all users (registered and unregistered) to:
  ○ View products.
  ○ Select items for purchase.
  ○ Register (new customer).
  ○ Login (existing customer).
• The following details of each product will be recorded:
  ○ A description of the product (each product will be recorded).
  ○ The price of the product.
  ○ The stock level of the product.
• The system will hold the following information about the customers:
  ○ The customer number (unique to each individual).
  ○ Name.
  ○ Address.
  ○ Telephone number.
  ○ Order history.
• The system will have different user roles and each role have specific jobs:
  ○ Admin – Will register a new staff.
  ○ Sales Assistant – Responsible for processing customer orders, entering of orders, taking payment, customer registration.
  ○ Invoice Clerk – responsible for sending out invoices to corporate customers.	

○ Sales Manager – access to all data (apart from the ability to register/edit a new Sales Manager). ○ Assistant Manager – Same as Sales Assistant and can also amend the price of products and have control over stock. ○ Stock Control Manager – Responsible for stock control. ○ Warehouse Assistants – Responsible for the dispatch of goods to customers. • The system will allow admins to register new members of staff and provide: ○ Username. ○ First name. ○ Second name. ○ Email. ○ Address. ○ Password. ○ Company name for the domain name. • All users require a User Id and Password. • The system will alert customers & staff when the stock of an item is low. • The system will have a section in the contact us page where they can query about stock availability. • The system implements payment gates: ○ Credit Card. ○ PayPal. • The system will have a loyalty program for discounts, based on the purchase history of the customer.	
GENERAL REQUIREMENTS OF THE SYSTEM ***(Continued)***	
Non-Functional Requirements: • The system will have information about the computing equipment stocked by the company. • The system needs to comply with relevant legislations such as data protection laws (GDPR) and online trading regulations. • The system must maintain compatibility with widespread legacy and contemporary desktop browsers. • The systems design must include a user-friendly interface, responsive layout, and visually appealing graphics consistent with the company's branding. • The system will include company logos, images and branding elements as part of the brand recognition.	

- The system will be compatible with various devices: desktops, laptops, tablets and smartphones to cater to diverse user preferences.
- The system will record sales for inventory management, financial reporting and customer service purposes.

The system will have security measure such as data encryption, secure payment gateways, and user authentication protocols will be implemented to protect sensitive information.

General Input Permissible:

User Authentication:

- Username -
  - Permissible Values: Characters from A-Z,a-z,0-9, underscores – **Data type:** strings.
  - Length: Minimum 6 characters, maximum 20 characters
- Password –
  - Permissible Values: any combination of characters & numbers – **Data type:** strings (after hashing).
  - Length: Minimum 8 characters, maximum 30 characters.
  - Additional Requirements: Must include one uppercase letter, one lower case letter, one digit and one special character. **This is VOID for testing**.

Product Creation:

- Product Title -
  - Permissible Values: Characters from A-Z,a-z,0-9, special characters (e.g., ., -, _) – **Data type:** strings.
  - Length: Minimum 1 character, maximum 50 characters.
- Product Price -
  - Permissible Values: positive decimal numbers to 2 decimal points (e.g., 3.2, 5.33, 7.0) – **Data type:** double.
  - Permissible Cost Range: Minimum £9.99, maximum £999.99.
- Product Stock Level -
  - Permissible Values: positive whole numbers (e.g., 9, 20, 99) – **Data type:** Integer.
  - Permissible Stock Range: Minimum 1, maximum 999.
- Product Category –
  - Permissible Values: Characters from A-Z,a-z,0-9 (e.g., Cars, Limited Sets, Tracks)– **Data type:** strings.
  - Length: Minimum 1 character, maximum 50 characters.
- Product Description –
  - Permissible Values: Character from A-Z,a-z,0-9 (e.g., small description of the product) – **Data type:** strings.
  - Length: Minimum 1 character, maximum 400 characters.
- Product Info –

- Permissible Values: Character from A-Z,a-z,0-9 (e.g., 1 car, 2 car 3 race tracks.) – **Data type:** strings.
- Length: Minimum 1 character, maximum 200 characters.

- Product Size –
  - Permissible Values: positive decimal numbers to 2 points (e.g., 1.45cm, 5.32cm.) – **Data type:** double.
  - Length: Minimum 0.99cm, maximum 99.99cm.
- Product Colour –
  - Permissible Values: Character from A-Z,a-z,0-9 (e.g., blue, orange, purple, blue and orange mix.) – **Data type:** strings.
  - Length: Minimum 3 character, maximum 50 characters.

User Creation:

- Personal Details – Name, Address, Email, Phone number, order history(**CUSTOMERS Only**).
  - Permissible Values: Characters from A-Z,a-z,0-9, special characters (e.g., ., -, _) – **Data type:** strings.
  - Length: Minimum 1 character, maximum 50 characters.
- Password –
  - Permissible Values: any combination of characters & numbers – **Data type:** strings (after hashing).
  - Length: Minimum 8 characters, maximum 30 characters.
  - Additional Requirements: Must include one uppercase letter, one lower case letter, one digit and one special character.
- Company Details – Employment Status and Job Role (Relevant only for **STAFF)**
  - Refer to **"Dropdown Menus"**.

User Permissions:

- Permissible Roles: Admin, Manager, Staff, Customer user
- Access Control:
  - Admin has full access.
  - Staff Assistant will have access to customer orders, details about the payment, and creating/modifying customers' details.
  - Invoice Clerk had access to invoices from corporate customers.
  - The Sales Manager has full access to data (cannot register/edit a new Sales Manager staff that falls to the Admin role).
  - Assistant Manager – Same access as a Sales assistant, but also has access to price of product and stock control.
  - Stock Control Manager – Full access over stock control.
  - Warehouse Assistants – Full access to shipping details of a product.

File Attachments:

- Permissible File Types: JPEG, PNG, PDF

- File Sizes: Maximum 5MB

Date and Time Selection:

- Permissible Date/Time Format: DD/MM/YYYY HH:MM:SS
- Additional Constraints: Time should be in 24-hour format.

Navigational Links:

- Permissible Options:
  - Each role will have a dedicated page, where they will have access and control over specific functionalities.
  - Everyone can access/navigate to “Home”, “Shop” , “About us”, “Contact us” pages.

Dropdown Menus:

- Permissible Options: Options pulled from the database and displayed as options in a dropdown list.

Checkbox Selection:

- Permissible Values: Checked or unchecked.

Search Queries:

- Permissible Values: Characters A-Z,a-z,0-9.
- Length: Minimum 1 character

Error Messages:

Login:

- Incorrect credentials.
- Username field is required.
- Password field is required

Register & Staff Creation:

- First name field is required.
- Last name field is required.
- Street field is required.
- City field is required.

- Post Code field is required.
- Email field is required.
- Password field is required.

Product Creation/Modification:

- Title field is required.
- Stock level field is required.
- Price field is required.
- Product info field is required.
- Product description field is required.
- Product size field is required.
- Product colour field is required.

Shopping page:

- Please select a product to add to basket.

Shopping cart page:

- Cannot place order, shopping cart empty.

Shipping Details Page:

- Incorrect details.

Viewing Orders:

- Cannot view orders: No order was selected.

Main System Interface: **Web Interface**.

System:	Speedo Models Company Website

Constraints:

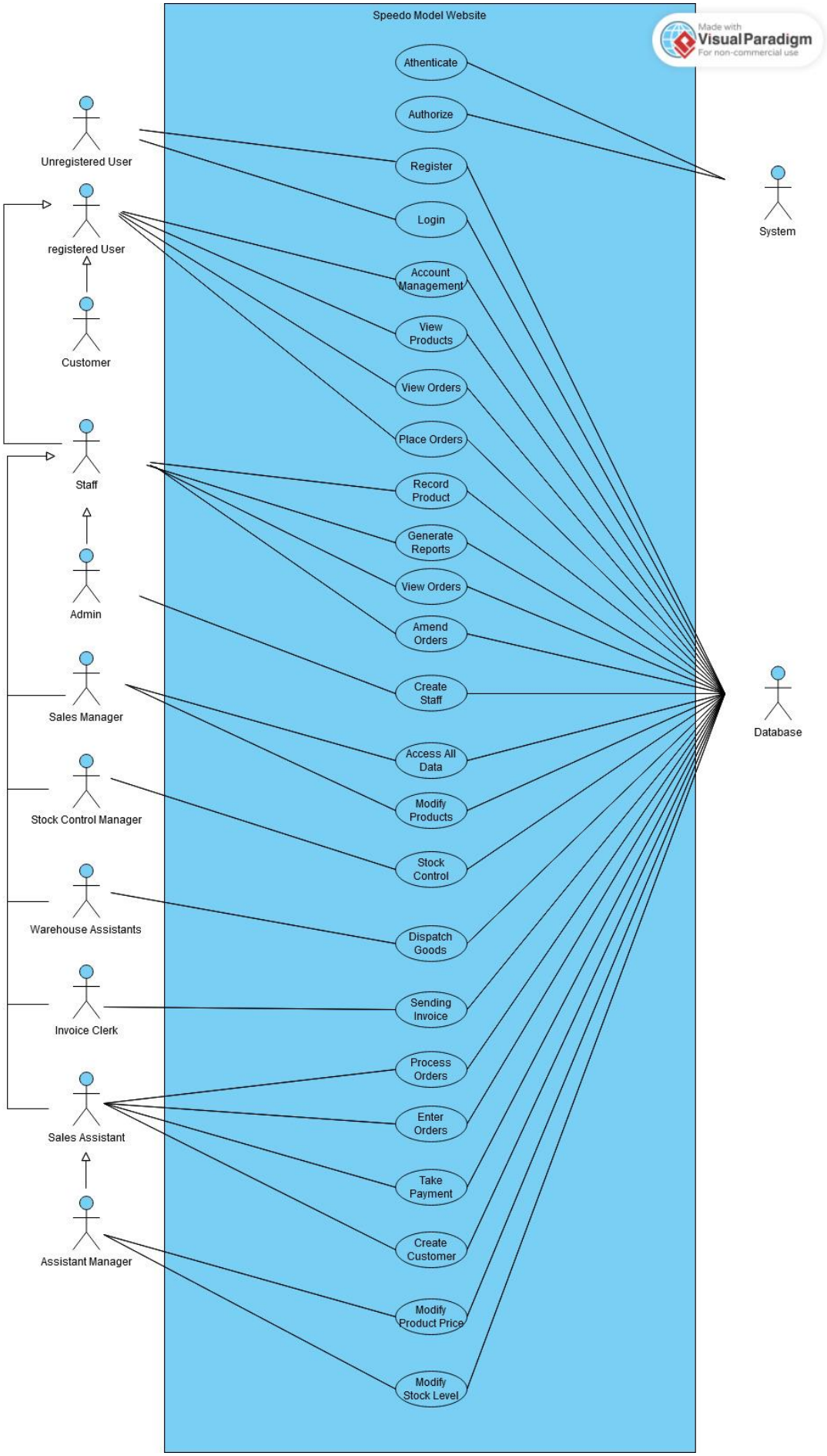
Constraint	Description	Implication
Project Deadline	The final project must be delivered by the 10th of June 2024. I must also provide 6 months of post-delivery support.	The deadline applies a timeline pressure on hitting project milestones within specific timeframes.
Budget Limitations	The project has a maximum fee cap of £15,000.	This budget constraint means the appropriate allocation of resources is essential and making sure I'm prioritizing specific features over others during the development stage.
Device and Browser Compatibility	The system needs to be compatible with various devices and maintain compatibility with a range of desktop browsers.	Design and development must ensure a responsive, cross-platform web application.
Potential International Expansion	Although the company is currently operating locally, there has been mention of plans for international expansion in the future.	The platform must be designed to accommodate multi-region support and scalability.
Social Media Integration Incorporated With User Registration/Login	The company has shown interest in allowing for registration/login via	The system will have to support social media API's and third-party authentication methods.

	Google and/or Facebook integrations.	
Measuring Customer Satisfaction	The company is interested in collecting feedback data from customers using methods such as conducting an online survey.	The system will need to include tools for collecting customer feedback, which will improve efforts to enhance user experience.

Each other constraint, from the third one onwards, must take into consideration the project deadline and budget limitations as possible implications.

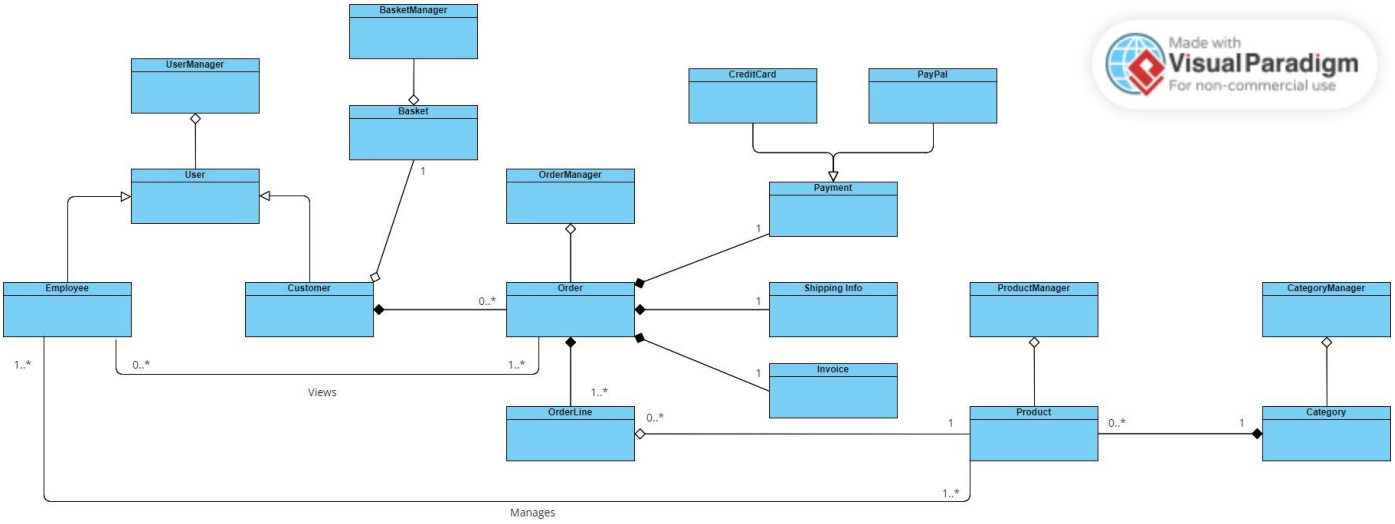
Initial Top Level Use Case Model

1.1.1.5:


Speedo Model Website
Made with Visual Paradigm For non-commercial use
Athenticate
Authorize
Register
Login
Account Management
View Products
View Orders
Place Orders
Record Product
Generate Reports
View Orders
Amend Orders
Create Staff
Access All Data
Modify Products
Stock Control
Dispatch Goods
Sending Invoice
Process Orders
Enter Orders
Take Payment
Create Customer
Modify Product Price
Modify Stock Level
Unregistered User
registered User
Customer
Staff
Admin
Sales Manager
Stock Control Manager
Warehouse Assistants
Invoice Clerk
Sales Assistant
Assistant Manager
System
Database

Initial Class Diagram

1.1.1.6:



Identification of resources and materials required

1.1.1.7:

Development Platforms

I have chosen to use the C# programming language, utilizing Visual Studio Community 2022 Integrated Development Environment (IDE) and the ASP.NET entity framework, with the ASP.NET Identity security system for authenticating and authorizing users and utilizing the bootstrap library for my UI design. This is my chosen technology stack.

Another technology I was thinking about using was Java as I am well versed with it. The backup plan was to use the Eclipse IDE as it is known for its configuration and customizability and integrate MySQL as the database with the use of the Spring Framework for both the MVC aspect and the utilization of Spring Security, which is a comprehensive security service which includes authentication and authorization. I looked

at utilising the flexible Tailwind CSS framework with Thymeleaf for the server-side, as the UI design stack.

C# & Technologies Background:

C# Is a modern, object-oriented programming (OOP) language, designed by Microsoft for building a wide range of applications on the .NET framework.

ASP.NET is a free web framework for C# and provides a robust environment for building dynamic web applications.

Entity framework is an open-source object-relational mapping (ORM) framework for ADO.NET, part of the .NET ecosystem. This technology is designed to enable developers to work with relational data using domain-specific objects, it supports three types of modelling workflow: database first, model first, and code first. I will be using the code first approach.

ASP.NET Identity is a membership system for building secure ASP.NET web applications, providing login functionality, user authentication and authorization.

Bootstrap will be used as the UI design stack. It is a free and open-source CSS and JavaScript framework used for responsive front-end development. Utilising different free design templates on the templates section of the Start Bootstrap website and/or paid themes on the themes section of the Get Bootstrap website, or free icons (**found in the resources section**).

I have been using this technology stack for the past few months, building robust websites for multiplatform uses.

C# & Technologies Concepts:

This technology stack emphasizes simplicity, robustness and versatility. It supports the MVC architecture, enabling clean data separation between application logic and the UI. It also offers features like LINQ for data manipulation and async/wait for asynchronous programming.

EF allows developers to work at a higher level of abstraction when dealing with data and its database. It supports LINQ queries, updating and schema migrations.

Identity includes profile support, OAuth integration and works with external login providers (e.g., Google, Facebook).

Bootstraps key concept is using a grid system for creating layouts and as discussed earlier, pre-designed templates, themes and other UI tools for quick and efficient web development.

C# & Technologies Advantages:

- The Visual Studio IDE offers very powerful debugging tools and very efficient and easy database integrations.
- There are strong security features and extensive libraries.
- Good performance and scalability.
- Extensive support and resources from Microsoft a multi-billion-dollar company.
- The utilization of Entity framework reduces the amount of boilerplate code required to access data and it's strongly typed, which helps to handle errors at the compile-time.
- ASP.NET Identity supports asynchronous programming patterns and integrates with EF for storing and managing user data.
- Bootstraps pre-designed templates allow for quick development, making it faster to create prototypes and to build web pages.
- Bootstrap's responsive and customizable design allows for creating websites that work on any device and that fit specific client needs.

C# & Technologies Disadvantages:

- It is primarily Windows-focused, even though the .NET core has cross-platform capabilities.
- It can be hard to learn the technology if your brand new to programming.
- Entity framework can introduce performance issues with the use of complex queries.
- ASP.NET Identity Configuring and customization of elements can be overwhelming for new developers.
- Bootstraps UI templates means very similar-looking websites can be found online, if the developer doesn't customize the website enough.

Java & Technologies Background:

Java is a highly popular, platform-independent Object-Oriented Programming language, which is known for its motto 'Write Once, Run Anywhere' which is a principle that exists thanks to the Java Virtual Machine (JVM), making it the ideal language for developing portable applications across various platforms and Operating Systems (OS's).

I was thinking of using Eclipse as the IDE, Eclipse is open-source and allows for a variety of languages thanks to its extensive plugin support.

I was thinking of using Spring as the framework, as it was developed to simplify the complexity of enterprise application development and it is more flexible than its

alternative – the Java EE standard. Since its development Spring has grown, supporting a wide range of application needs such as web development, security, and data access. Spring Security is a highly customizable authentication and authorization control framework.
I was thinking of utilizing Tailwind CSS for the UI. It is a framework used for creating custom designs. In conjunction with Tailwind, I was looking at using Thymeleaf, which is a server-side Java XML/HTML5 Template engine that is suited for building web environments; however, it's not restricted to this.

Java & Technologies Concepts:
Java is class-based, allowing developers to create modular programs (with concepts such as data separation) and reusable code. Java applications are compiled to bytecode, which can run on any JVM, regardless of the underlying architecture.
Eclipse IDE is built on a modular plugin architecture, which allows for different development tools and languages. Main features include covering a large scope of coding, debugging, and testing tools specific to Java.

The core concept behind Spring is the Inversion of Control (IoC), which helps in decoupling application components. Other concepts include Aspect-Oriented Programming (AOP) which promotes modular code through component separation and MVC (Model, View, Controller) Separating data into three connected components: Model, which deals with business logic. View which is responsible for rendering the model data, usually via an HTML output that the client's browser can interpret. Controller: Handles user's request, works on the model data, and returns a view to the user (one of the most important components of the MVC architecture).

Spring Security provides different security features: Authentication, authorization, protections against common vulnerabilities, and session management.
Tailwind provides classes that allow developers to build complex designs directly within the HTML markup. Thymeleaf processes HTML files on the server, placing data into templates before it is sent to the client.

Java & Technologies Advantages:

- Java is cross-platform, which enhances flexibility and reduces development and deployment costs.
- Strong memory management, including automatic garbage collection.
- Like mentioned above the Eclipse IDE is highly customizable and supports an array of plugins. It also has a strong community of users behind it.
- The Spring framework can fluidly integrate with ORM (Object Relational Mapping) tools.
- Spring Security integrates well with the Spring ecosystem and is highly customizable to fit security requirements.
- MySQL database is scalable as it is capable of handling large volumes of data.
- The discussed UI technology stack allows for design flexibility, as Tailwind's utility classes offer really good customization along with rapid prototyping capabilities.

Java & Technologies Disadvantages:

- Compared to C#, Java has a slower runtime for certain types of web applications.
- Eclipse has a vast array of complex features and plugins which can be overwhelming for beginners.
- Configuring the Spring Framework can be a complex task, especially in large projects or those with abundant XML configurations.
- For Spring Security, the intensity of the system might introduce complications in smaller projects.
- Due to the nature of MySQL being able to handle large datasets, this can lead to performance issues in scenarios with large databases.
- The discussed UI technology stack has a steep learning curve, requiring more time, and therefore it will have extra costs.

Justification/Choice of Development Platform

When it comes to building robust and secure web applications, C# offers a more compelling set of advantages. The integrations of the surrounding technology: Entity framework with the code-first approach, ASP.NET Identity with bootstrap and features such as auto-completing code and pre-written code configurations that are easily accessible and editable (for example, overriding methods), coupled with personally having more experience working with this technology stack, makes it an easy choice for me.

Compared to the technology stack described with Java, where a lot of the web-based application features are not pre-written and so will require starting from scratch, building everything from the ground up, which will lead to me as the developer proposing a higher budget for development due to the increased difficulty level and time-consumption using this technology stack would cost – not to mention potential

performance issues due to the nature of java's slower runtime environment when it comes to web development in comparison to C# and it's technology stack.
Although Java is a viable option and the discussed technology stack that comes along with it has a variety of advantages, overall, as an entry-level developer, Visual Studios C# ASP.NET is a more efficient & effective solution for me, because I have more experience in using it.

Conclusion

To conclude this report has allowed me to collect enough evidence to use when creating the application and I will use it as a reference point in the design, implementation process, and beyond. With this clear vision and plan, I have positioned myself to develop a highly functional platform that will help Speedo Models operate and engage with its customers on a more efficient and effective level. The valuable insights collected during the planning stage set up the remaining stages for a successful project outcome that will help Speedo Models stand out from their competition.

Identification of Information sources to be used:

1.1.1.8:

Appendices & Resources

Initial Class Diagram (Further research for understanding):
https://www.youtube.com/watch?v=q8UbnNsbVYk

HTML all the things podcast – for further understanding of dealing with clients and their expectations from a web development standpoint:
https://open.spotify.com/show/2MWqU5ZbO69jy3RZ74wgdM

The Hawthorne Effect:
https://en.wikipedia.org/wiki/Hawthorne_effect

GDPR:

https://gdpr-info.eu/

Constraints:

https://www.mobindustry.net/blog/what-are-constraints-in-a-software-development-project/

Development Platforms – Background:

C#:

https://learn.microsoft.com/en-us/dotnet/csharp/tour-of-csharp/

Visual Studio Community 2022:

https://visualstudio.microsoft.com/vs/community/

ASP.NET:

https://learn.microsoft.com/en-us/aspnet/overview

Entity Framework:

https://www.entityframeworktutorial.net/

https://learn.microsoft.com/en-us/dotnet/csharp/linq/get-started/introduction-to-linq-queries

ASP.NET Identity:

https://learn.microsoft.com/en-us/aspnet/identity/overview/getting-started/introduction-to-aspnet-identity

Bootstrap & Design templates:

https://getbootstrap.com/

https://startbootstrap.com/

Java:

https://beginnersbook.com/java-tutorial-for-beginners-with-examples/

Eclipse IDE:

https://www.eclipse.org/ide/

Spring Framework:

https://spring.io/projects/spring-framework

Java EE Standard:

https://www.oracle.com/uk/java/technologies/java-ee-glance.html

Spring Security:

https://spring.io/projects/spring-security

MySQL:

https://www.youtube.com/watch?v=7S_tz1z_5bA

https://www.mysql.com/

Thymleaf and Tailwind CSS:

https://www.wimdeblauwe.com/blog/2020/02/17/using-tailwind-css-with-spring-boot-and-thymeleaf/

https://tailwindcss.com/

https://www.thymeleaf.org/

C# vs Java:

https://softteco.com/blog/java-vs-csharp

Bibliography

References:

1.1.1.2 Information gathered to clarify the brief:

- LinkedIn. *Face-to-Face Interviews – Advantages and Disadvantages*. Available at: https://www.linkedin.com/pulse/face-to-face-interviews-advantages-disadvantages-charlie-marshall/ (Accessed: 16/02/24)

Initial Class Diagram:

- YouTube. *UML Class Diagram for E commerce Website*. Available at: https://www.youtube.com/watch?v=q8UbnNsbVYk (Accessed: 09/02/24)